

Limb-segment volume from phone photos

Four versions of the pipeline, what went wrong, and what fixed it

Mean error between the two bands, six captures with ground truth

31% → 5.7%

v3 (cut at Stage 5), sound runs only

v4, today (padded photos)

Four versions of the pipeline

Same idea throughout: 8 phone photos + 10 cm ArUco cube → VGGT 3D points → mesh → volume

v1 Old

main branch

VGGT centre-crops each photo

Floor removal, DBSCAN

Cut the point cloud at the marker

Alpha / Poisson mesh

Volume: watertight, or voxel

flood-fill when the mesh leaks

One band only: measures below it

v2 Reworked + Stage 0

12–24 Aug

Stage 0: find cube, limb, bands

(GroundingDINO + SAM), gate

bad photos, crop around them

Learned band colour

Floor, marker, cut-rule fixes

Poisson with alpha fallback

Web review before the cut

v3 Cut at Stage 5

31 Aug

Cut the watertight mesh, not
the point cloud

Two bands: measure between them

Reviewer places the cut on a
surface in the web app

Cohort 0–6 + test captures

measured against water / tape

v4 Current

18–21 Sep

Second, wider MLS pass

Ghost-voxel spacing bug fixed

Projected band plane as backup

Pad the whole photo; Stage 0

no longer crops

Marker-scale check removed

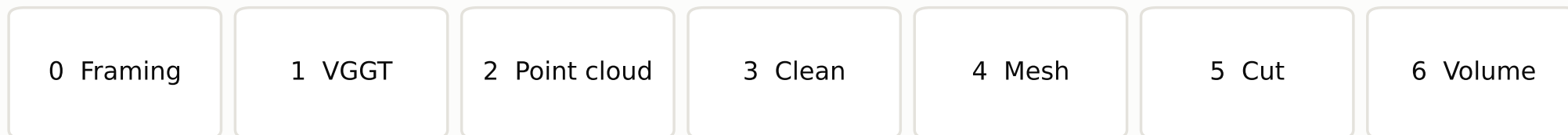
TSDf fusion: optional, off

Limb skeleton: smooth rings

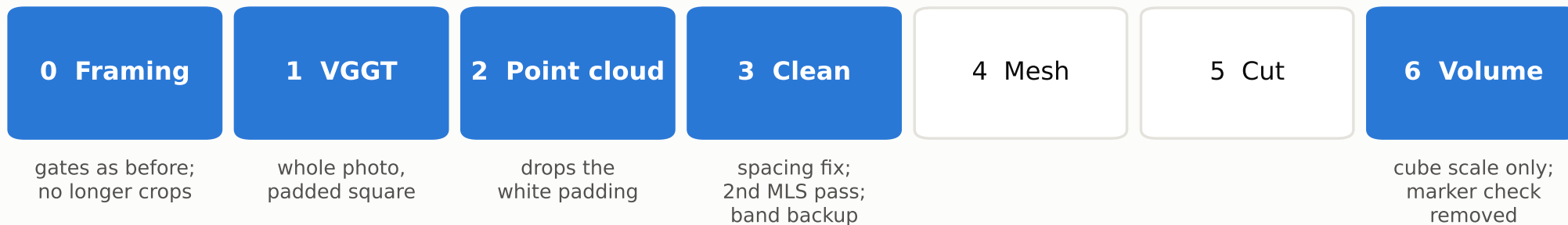
From v3 to v4 at a glance

Same seven stages; blue marks what changed

**v3 reworked
(cut at Stage 5)**



v4 current



v3 Stage 0 cropped each photo to a square that slid to fit the cube and bands.

That moved the optical centre off the middle, and VGGT distorted the whole scene.

v4 hands VGGT the whole photo instead. That one change is most of the gain.

From v3 to v4, stage by stage

Only the stages that changed; Stages 4 and 5 are the same code

Stage	v3 (reworked)	v4 (current)	Why
0 Framing	Find cube, limb, bands; gate; crop a sliding square around them	Find, gate, learn band colour as before; pass the photo whole	The slide put the optical centre 19-74 px off the middle
1 VGGT	Gets Stage 0's 518 px crop	Gets the whole photo, shrunk and padded to 518 px	VGGT assumes the centre is the middle of the image
2 Point cloud	Every confident pixel	Also drops the white padding	Padding has no geometry
3 Clean	Ghost voxel sized by a buggy spacing; one MLS pass (4x); lost band if colour fit failed	Spacing fixed; 2nd MLS pass (16x) merges the double skin; projected plane as backup	Poisson closes instead of the alpha wrap; bands no longer vanish
6 Volume	Cube-volume scale	Cube-volume scale (marker check tried, removed)	Ruler agreed with the cube

Each v4 change, and what it did

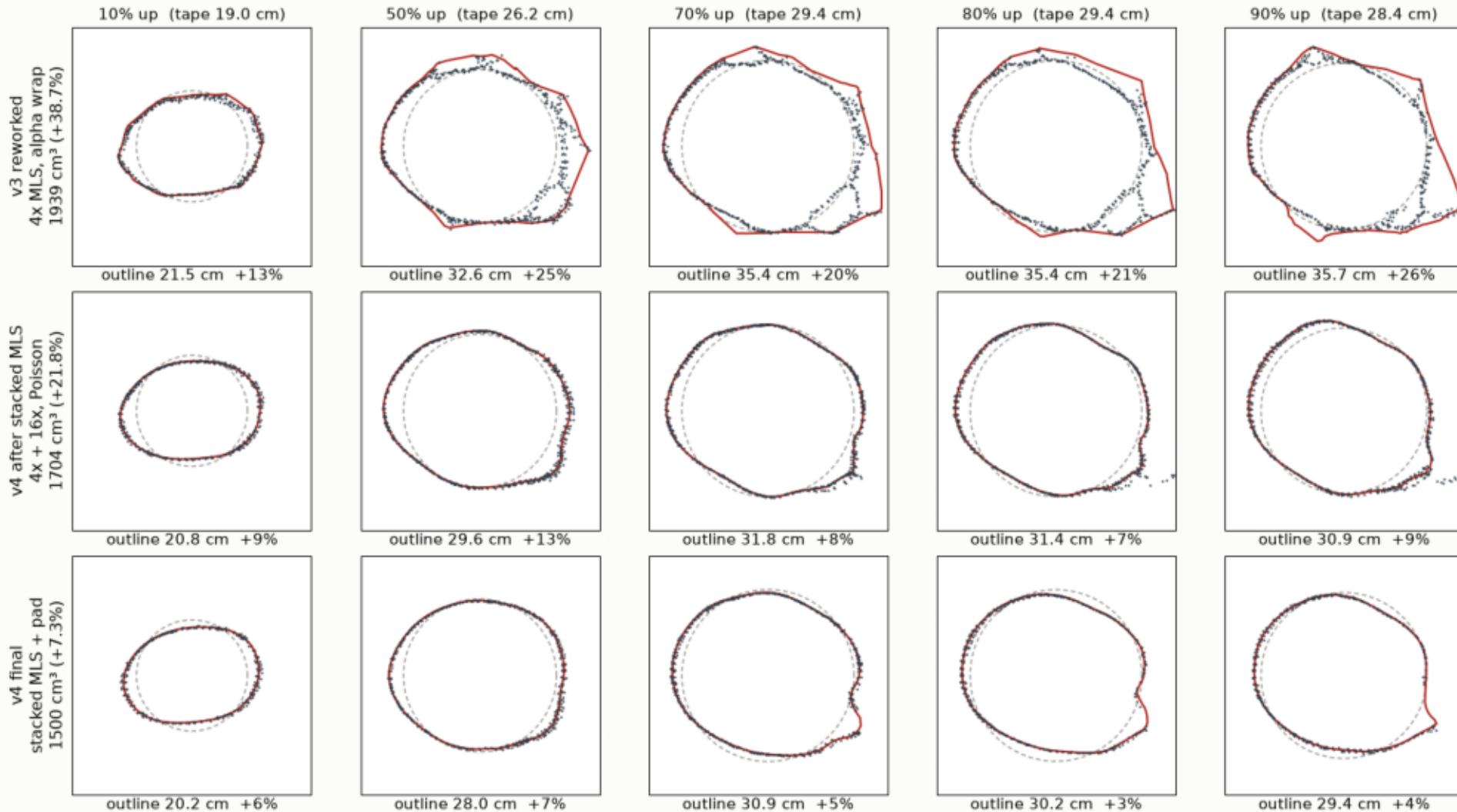
Volume between the bands against tape / water; changes applied in order

Change	test6 (tape)	test5 (water)	Cohort (water)
v3 starting point	+38.7% (alpha wrap)	+24.9% (alpha wrap)	+25 to +36%; 2_left +222%
+ 2nd MLS pass	+21.8% Poisson closes	+17.8% Poisson closes	within 1% of v3 where Poisson already closed
+ ghost-voxel spacing fix	+21.8% (unchanged)	—	stops denser photos being cleaned coarser
+ band-plane backup	unchanged	unchanged	0_left 10 cm ³ → −1.5% 2_left +222% → +37%
+ pad, do not crop	+7.3%	+4.7%	−5.5% to +11.2% mean error 29% → 5.7%

The MLS pass and the two bug fixes made the mesh sound; padding removed the systematic bias.

Ring gallery: test6 cross-sections

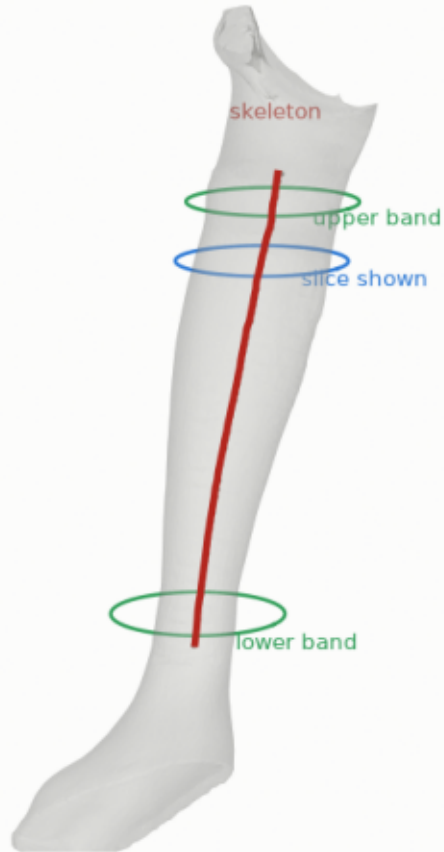
Dark dots = cleaned points, red = mesh outline, grey dashed = tape; same 13 cm window everywhere



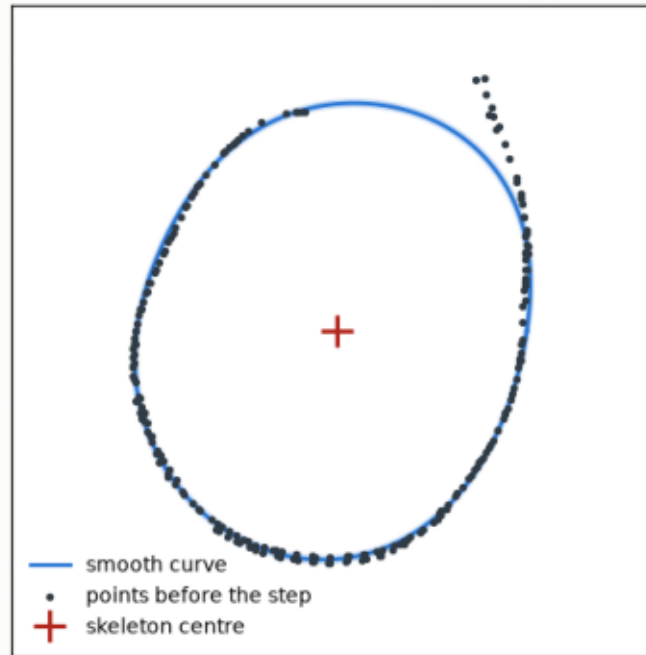
Limb skeleton: how it works

Centre of each 0.5 cm slice, smoothed up the leg; each ring fitted with one smooth curve

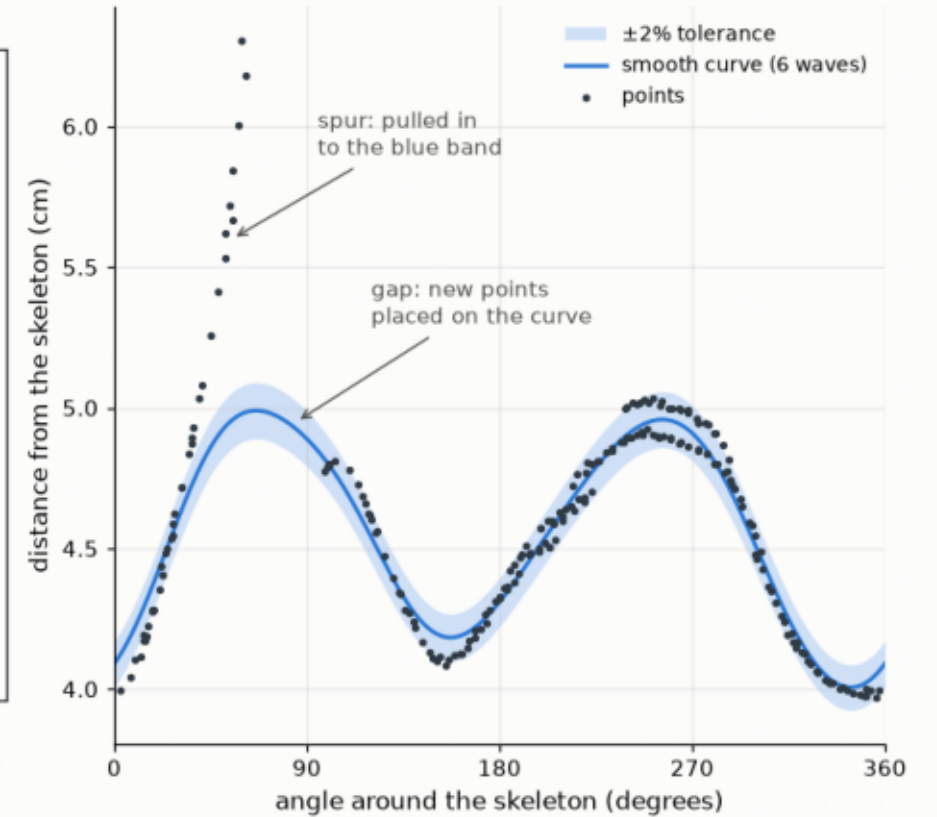
Leg mesh with its skeleton



One slice from above (85% up)

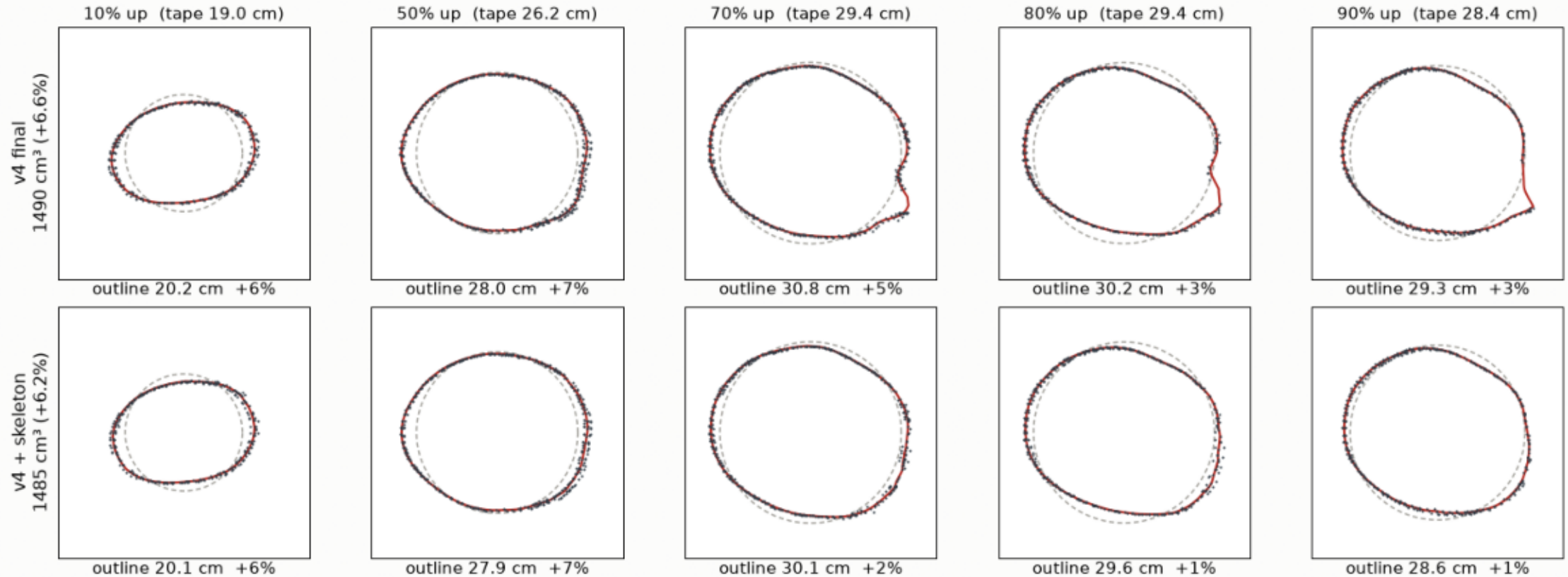


The same slice unrolled



Limb skeleton: smoothing each ring

One smooth curve per slice: spurs moved in, dents moved out, gaps filled. Dots = points, red = mesh, dashed = tape



Limb skeleton: effect on volume

Stages 3-6 rerun on the padded runs, step off and on; volume between the bands

Capture	Truth (cm ³)	Skeleton off	Skeleton on
test6	1398.6 tape	1490.4 +6.6%	1485.2 +6.2%
test5	2070 water	2165.8 +4.6%	2167.6 +4.7%
0_right	1830 water	1734.5 -5.2%	1743.5 -4.7%
1_left	1600 water	1612.0 +0.8%	1591.1 -0.6%
6_left	2800 water	2648.8 -5.4%	2647.2 -5.5%
2_left	1050 water	1179.4 +12.3%	1180.2 +12.4%
Mean error		5.8%	5.7%
Mean error (bias)		+2.3%	+2.1%

Shape is cleaner (test6 upper rings: outline error +3-5% → +1-2%); volume barely moves.
Nothing got worse, so the step is now on by default (LIMB_SKELETON=on).

Problems we hit, and how we solved them

v3 → v4, measured against tape and water at each step

Problem	Cause we found	Fix	Effect (measured)
Alpha fallback inflated the volume	Double 'ghost' skin 1 cm apart; Poisson tunnels, falls back	2nd MLS pass (16x) merges the two skins	test6 +38.7% → +21.8% test5 +24.9% → +17.8%
Sharper photos read bigger	Cleaning spacing measured on a sample against itself	Measure the spacing on the whole cloud	Full-res photos no longer cleaned coarser
A band vanished, sliver volumes	Colour-fitted plane gated out, nothing to fall back on	Keep Stage 0's projected plane as a backup	0_left 10 cm ³ → −1.5% 2_left +222% → +37%
Leg 9% too big against the cube	Stage 0's crop moves the optical centre 19–74 px	Pad the whole photo; do not crop	Mean error 29.0% → 5.7% (6 captures)
Two scales disagree	Marker corners unreliable on busy texture	Trust the cube volume; check removed	Ruler agrees with the cube scale
Reflective floor: a 15 L 'can'	Floor fit fails on the mirror sheen	Shoot on the matte tile (capture rule)	5_right, fanta_red excluded

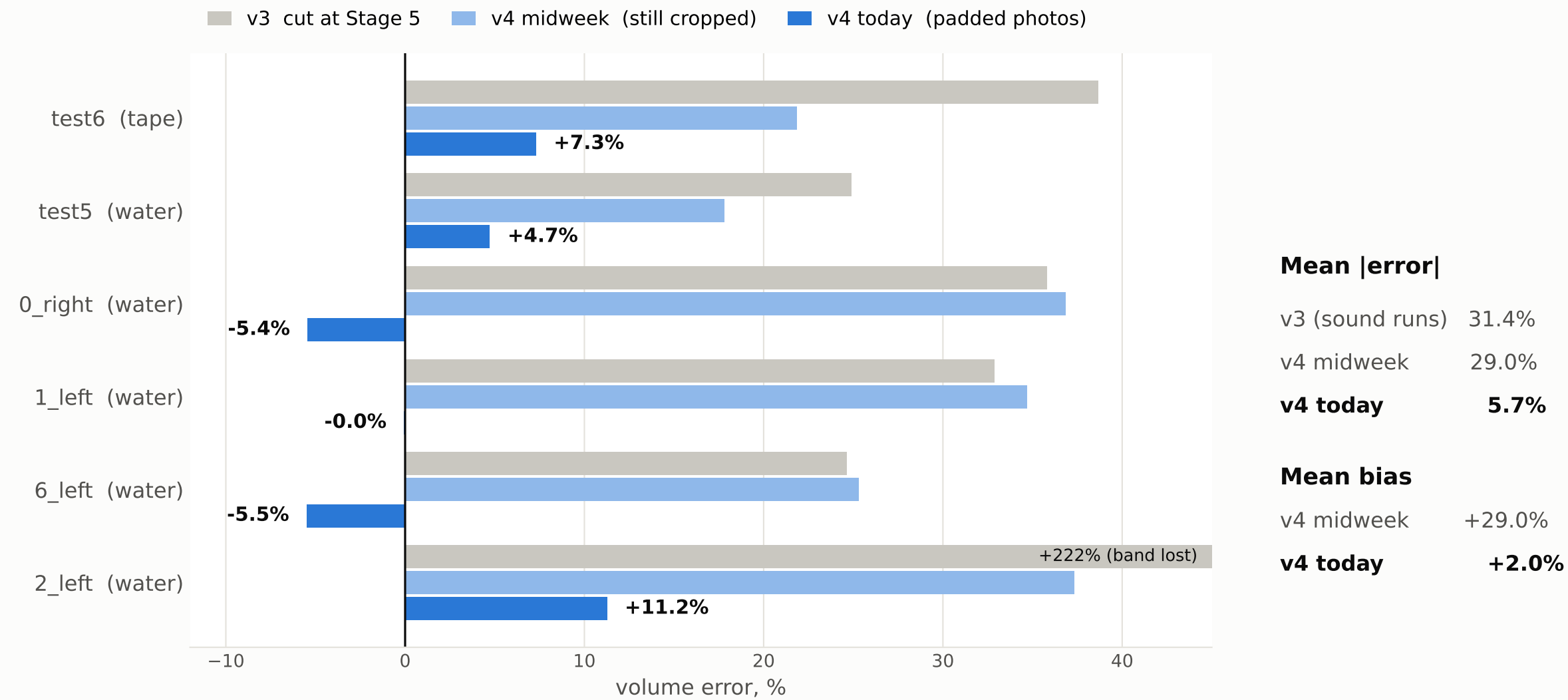
What we ruled out on the way

Each was tested with a measurement, not argued

Suspect	Test	Verdict
Cube size / scale	Cube 10.0 cm, markers 5.0 cm, measured; scale checked against markers	Not the cause
Camera lens (FOV)	VGGT focal vs iPhone EXIF; worst on the can, which reads right	Not the cause
Photo resolution	HEIC originals; VGGT at 1022 px	Worse, or breaks
The model itself	MapAnything, MoGe-2, Meshy AI on the same photos	Same bias, or unusable
How points are merged	TSDf fusion; inner ghost sheet; silhouette carving	Cleaner cube, same leg
Leg shake, hair, cut position	Four capture styles; hairless leg; band colour vs plane	Not the cause
Stage 0 off-centre crop	VGGT on original photos, padded or centre-cropped	The cause: about half the error

Volume between the bands: error against ground truth

Full pipeline, end to end. Bars left of zero read low, right of zero read high.



Full results table: volume between the bands (cm³)

Truth: tape disc model (test6) or water displacement (others)

Capture	Truth	v3 (cut at Stage 5)	v4 midweek	v4 today (padded)
test6	1399 (tape)	1939 +38.7%	1704 +21.8%	1500 +7.3%
test5	2070 (water)	2585 +24.9%	2439 +17.8%	2168 +4.7%
0_right	1830 (water)	2485 +35.8%	2504 +36.8%	1731 -5.4%
1_left	1600 (water)	2126 +32.9%	2155 +34.7%	1599 -0.0%
6_left	2800 (water)	3490 +24.6%	3508 +25.3%	2647 -5.5%
2_left	1050 (water)	3381 +222.0% band lost	1442 +37.3%	1168 +11.2%
Mean error		63.2% (31.4% without 2_left)	29.0%	5.7%
Mean bias		+63.2%	+29.0%	+2.0%

Band circumference against the tape (cm)

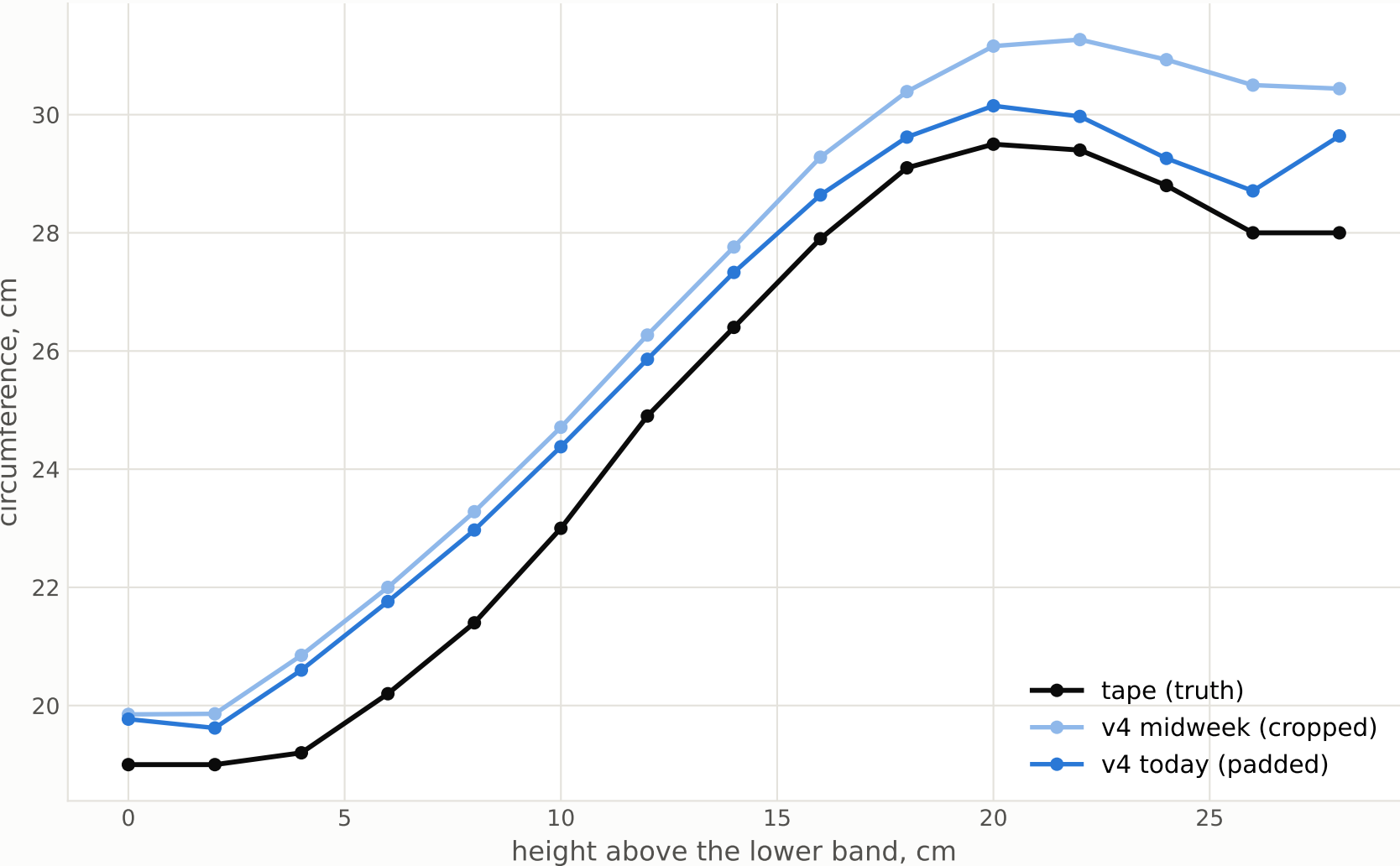
Lower band / upper band, measured at the cut planes

Capture	Tape lower / upper	v4 midweek (cropped)	v4 today (padded)
test6	19.0 / 28.0	19.95 (+5%) / 31.85 (+14%)	19.61 (+3%) / 29.42 (+5%)
test5	22.0 / 32.5	22.16 (+1%) / 34.55 (+6%)	22.19 (+1%) / 32.86 (+1%)
0_right	20.5 / 29.5	19.98 (−3%) / 33.30 (+13%)	18.59 (−9%) / 29.45 (0%)
1_left	21.5 / 32.0	20.80 (−3%) / 33.95 (+6%)	19.89 (−7%) / 30.13 (−6%)
6_left	24.3 / 35.0	23.69 (−3%) / 39.10 (+12%)	22.15 (−9%) / 34.99 (0%)
2_left	18.5 / 27.5	19.05 (+3%) / 30.61 (+11%)	18.69 (+1%) / 27.52 (0%)

The upper band is now within 1% on four of six captures. The lower band reads a few percent low on the cohort; tape repeatability measured on test5 was 0.8 cm on average (about 4%).

test6: circumference every 2 cm along the leg

Tape measured every 2 cm; the pipeline sliced at the same heights



test6	Truth	Midweek	Today
Length, cm	27.5	29.92 +8.8%	28.87 +5.0%
Girth, mean	tape	+5.2%	+1.7%
Volume	1398.6	+21.8%	+7.3%

What remains on test6 is mostly length (band separation +5%).

Open question: was the ruler read cord centre to centre, or between inner edges? One cord $\approx$ 3.6%.

v1 and v2: the numbers that exist

They measured different quantities, so they are not in the main table

Comparison	Quantity	Captures	v1 Old	v2 / v3 Reworked
README set	below one band	5 older captures	22.17%	3.29% (re-run 3.12%)
Cohort, 4 Sep	foot to upper band	0_left, 0_right, 1_left, 1_right	11.8%	34.4% (v3)
test5, 18 Sep	old cut: 3 markers, 40.8 cm	test5	not comparable	—

- v1 cannot cut between two bands; it only measures below one
- On the cohort, v1's 11.8% came mostly from leaky voxel flood-fill meshes (3 of 4 runs)
- The Stage 0 crop, fixed today, is the largest known reason v3 trailed v1 on the cohort
- v4 has not yet been re-run on the README set or in below-one-band mode

Open items

- Re-run the can control: its cube fell to the alpha fallback on today's run
- Ruler definition on test6: cord centre to centre, or inner edges?
- Re-run the README set and the below-one-band mode on v4
- Remaining cohort captures (0_left, 1_right, 2_right, 3-6) on v4
- Capture rules: matte floor, cube and both bands in every photo
- Deferred, lower priority now: fine-tuning VGGT; training a MoGe-VGGT join